ABC TECHNOLOGIES - SOC SIMULATION PROJECT

PROJECT PROPOSAL

Project Title: Enterprise Security Monitoring and Incident Response Simulation
Date: 7th August, 2026

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Organization: ABC Technologies Ltd.

1. Executive Summary

This project proposal outlines the implementation of a comprehensive Security Operations Center (SOC) simulation for ABC Technologies Ltd., a fictional medium-sized enterprise with 250 employees across two locations in Ghana. The project aims to demonstrate the full incident response lifecycle through the deployment of industry-standard security tools including Splunk SIEM, Suricata IDS, and Python automation.

The simulation will provide hands-on experience in:

- Security monitoring and alert triage
- Threat detection and analysis
- Incident response and management
- Security automation and reporting
- Risk assessment and compliance

Project Duration: 6-8 weeks

2. Project Overview

2.1 Background

ABC Technologies Ltd. is a growing IT services company with operations in Accra and Kumasi. Recent security incidents, including failed login attempts, malware detections, and suspicious network traffic, have highlighted the need for a dedicated security monitoring capability.

2.2 Objectives

- Deploy a virtual SOC environment for security monitoring
- Implement Splunk Enterprise for log aggregation and analysis
- Deploy Suricata IDS for network intrusion detection
- Develop automated incident response procedures
- Create comprehensive security documentation

2.3 Scope

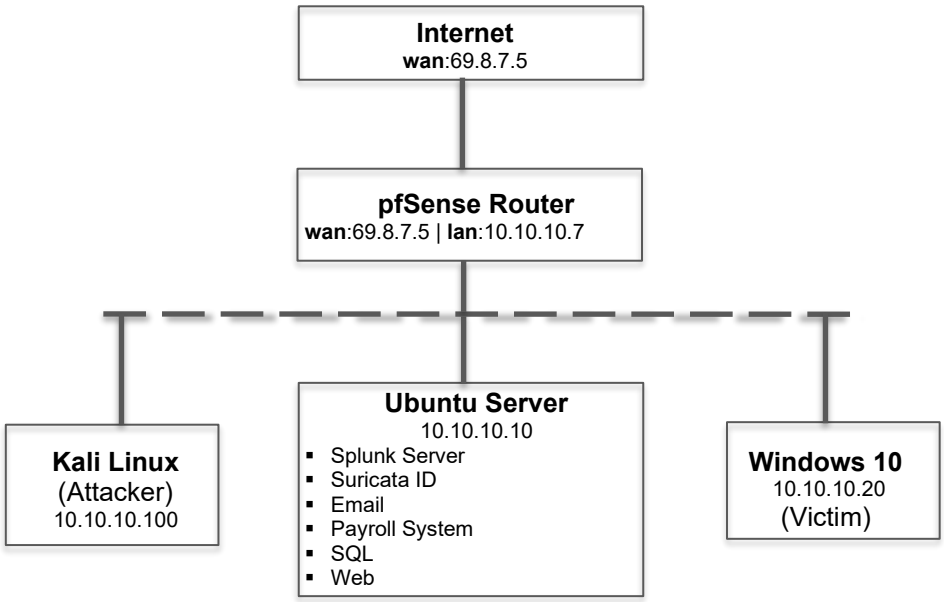
In-Scope:

- Deployment of virtual lab environment
- Installation and configuration of Splunk SIEM
- Implementation of Suricata IDS
- Development of security monitoring dashboards
- Creation of incident response playbooks
- Production of executive-level reports

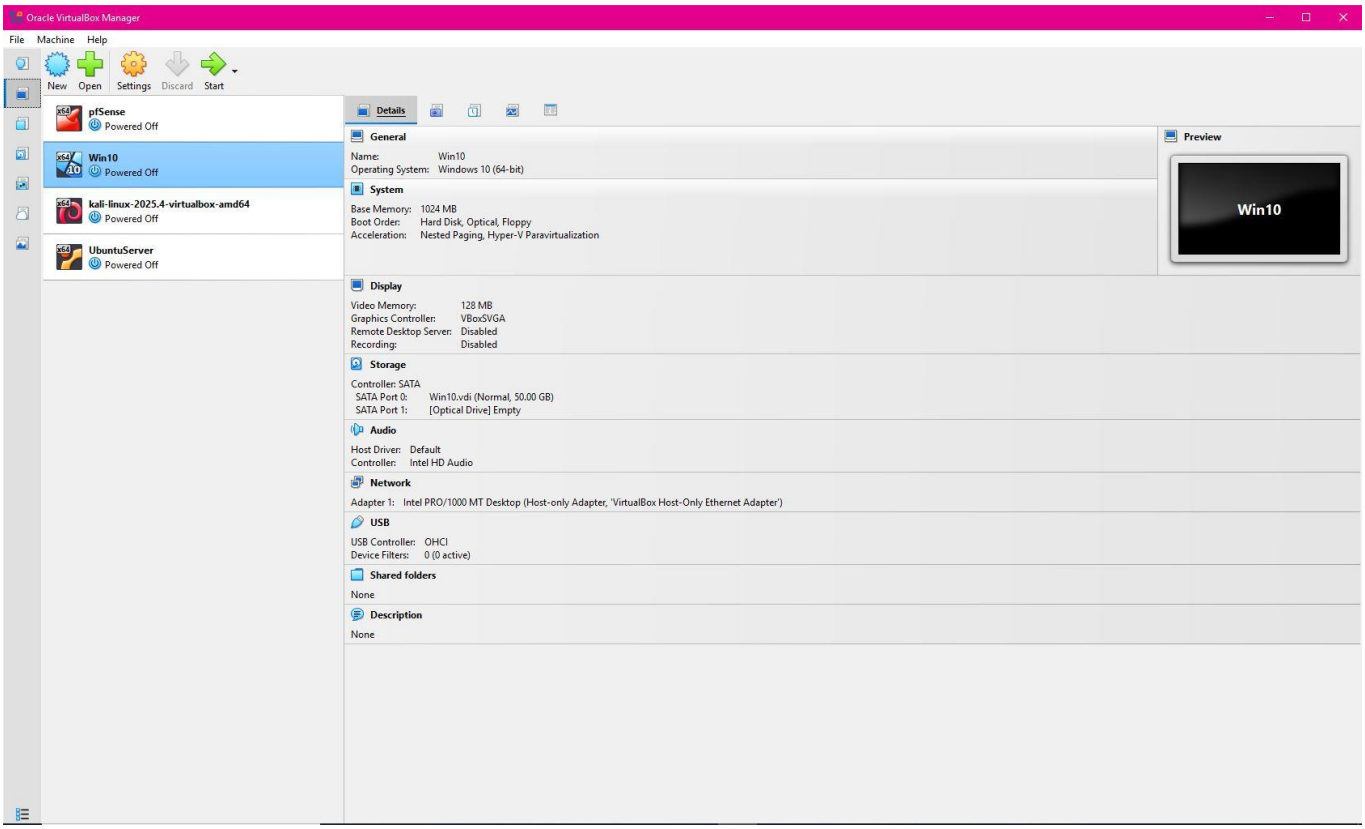
Out-of-Scope:

- Physical security controls
- Cloud infrastructure deployment
- Production environment implementation

3. Technical Architecture



3.1 Virtual Lab Infrastructure



3.2 Technology Stack

Component	Technology	Purpose
Operating Systems	Ubuntu Server, Kali Linux, Windows 10	Lab environment
Virtualization	VirtualBox	Virtual machine management
SIEM	Splunk Enterprise	Log aggregation and analysis
IDS/IPS	Suricata	Network intrusion detection
Packet Analysis	Wireshark	Network traffic inspection
Database	SQLite	Data storage
Version Control	Github	Source code management

4. Project Phases

Phase 1: Planning and Design (Week 1)

- Define project scope and objectives
- Design network architecture
- Create project timeline
- Identify resource requirements

Phase 2: Environment Setup (Week 2)

- Install and configure Ubuntu Server
- Set up Kali Linux attacker machine
- Deploy Windows 11 victim machine
- Install and configure Splunk Enterprise
- Deploy Suricata IDS
- Configure network connectivity

Phase 3: Configuration and Integration (Week 3)

- Configure Splunk data inputs
- Set up Suricata detection rules
- Create Splunk dashboards
- Implement log forwarding
- Configure alerting

Phase 4: Attack Simulation (Week 4)

- Execute port scanning (Nmap)
- Perform brute force attacks (Hydra)
- Simulate malware (EICAR test file)
- Conduct ping sweeps
- Test privilege escalation

Phase 5: Incident Response (Week 5-6)

- Monitor and analyze alerts
- Investigate security events
- Contain and remediate incidents
- Document findings
- Create incident reports

Phase 6: Reporting and Documentation (Week 7-8)

- Create risk assessment
- Develop incident response playbooks
- Generate executive reports
- Prepare final presentation

5. Deliverables

Deliverable	Description	Due Date
Project Charter	Project scope and objectives	Week 1
Technical Architecture	Network diagram and specifications	Week 1
Installation Guide	Step-by-step setup documentation	Week 2
Risk Assessment	Asset valuation and risk analysis	Week 3
Incident Response Playbook	Standard operating procedures	Week 6
Security Incident Report	Attack analysis and response	Week 7
Executive Summary	Management-level report	Week 8
Final Presentation	Project overview and demonstration	Week 8

6. Risk Management

Risk	Likelihood	Impact	Mitigation
Virtual machine resource constraints	Medium	High	Allocate sufficient host resources
Configuration errors	High	Medium	Document and test configurations
Limited Splunk data ingestion	Low	High	Monitor daily usage
Security tool conflicts	Medium	Medium	Test in isolated environment

7. Timeline and Milestones

Week	Action
1	Project Planning
2	Environment Setup
3	Configuration & Integration
4	Attack Simulation
5-6	Incident Response
7-8	Reporting & Documentation

Key Milestones

- [ ] Virtual lab operational (Week 2)
- [ ] Splunk receiving logs (Week 3)
- [ ] Suricata generating alerts (Week 3)
- [ ] First attack simulation (Week 4)
- [ ] Incident response execution (Week 5)
- [ ] Final report completed (Week 8)

8. Project Team

Role	Name	Responsibility
Project Lead	Christian Zigah	Overall coordination
SOC Analyst		Monitoring and analysis
Systems Administrator		Infrastructure management
Security Engineer		Tool configuration
Compliance Officer		Documentation and reporting

9. Budget and Resources

Resource	Specification	Estimated Cost
Virtual Machines	3 VMs (4GB RAM each)	Free (VirtualBox / Vmare)
Splunk Enterprise	Free license (500MB/day)	Free
Suricata IDS	Open source	Free
Kali Linux	Open source	Free
Linux Server	Open source	Free
Windows 10 lite	Trial Version	Free
Development Time	200 hours	GH¢ ----------

10. Success Criteria

- [ ] Install all Virtual Machines and all Assets running successfully.
- [ ] Splunk successfully ingests logs from all sources
- [ ] Suricata detects simulated attacks
- [ ] Security dashboards provide actionable insights
- [ ] Incident response procedures executed successfully
- [ ] Comprehensive documentation completed
- [ ] Executive-level reports delivered

11. Approval Signatures

Role	Name	Signature	Date
Project Sponsor	_____	_____	_____
Project Lead	_____	_____	_____
Security Manager	_____	_____	_____